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The effects of oxygen fugacity and sulfur on the pressure of vapor-saturation of magma

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Abstract

Geobarometers are commonly used to determine the pressure (and hence depth) of magmatic bodies. For instance, at equilibrium, the concentration of dissolved volatiles in a vapor-saturated melt can be used as a barometer: this is the pressure of vapor-saturation (P^v_{sat}). Most determinations of P^v_{sat} assume that melt and vapor contain only oxidized C-O-H species. However, sulfur is the third most abundant volatile element in magmas, and oxygen fugacity (f_{O_2}) exerts a strong influence on the speciation of the melt and vapor. To explore how S and f_{O_2} affect calculations of P^v_{sat}, we model a Hawaiian tholeiite that contains both reduced and oxidized C-O-H-S species in the melt and vapor. We find that excluding reduced C-O-H species in the system can result in significant underestimations of P^v_{sat} under reducing conditions (ΔFMQ < 0). The effect of S on P^v_{sat} is small except in the vicinity of the "sulfur solubility minimum" (SS^{min}; 0 < ΔFMQ < +2), where excluding S-bearing species can result in underestimates of P^v_{sat}.

The implications of these results depend on the volatile concentration of the system being investigated, its f_{O_2}, and the melt composition and temperature. Our results suggest there will be little impact on P^v_{sat} calculated for mid-ocean ridge basalts because their f_{O_2} is above where reduced C-O-H species become important in the melt and vapor and yet below the SS^{min}. However, the f_{O_2} of ocean island and arc basalts are close enough to the SS^{min} and their S concentrations high enough to influence P^v_{sat}. However, high-CO_2 and high-H_2O concentrations are predicted to reduce the effect of the SS^{min}. Hence, P^v_{sat} calculated for shallowly trapped melt inclusions and matrix glass are more affected by the SS^{min} than deeply trapped melt inclusions. Lunar and martian magmas are typically more reduced than terrestrial magmas, and therefore accurate P^v_{sat} calculations for them require the inclusion of reduced C-O-H species.

Keywords: Pressure of vapor saturation, thermodynamics, oxygen fugacity, carbon, hydrogen, sulfur; Experimental Halogens in honor of James Webster

Introduction

The concentrations of dissolved volatiles (e.g., H_2O, CO_2) in the melt phase of magma are widely used to determine total pressure (P) (e.g., Anderson et al. 1989; Blundy and Cashman 2008). Such determinations are based on analyses of volatile components in quenched glasses coupled with the assumption that the melt from which the glass formed was in equilibrium with a vapor prior to quenching. The basis of this approach is that for a given temperature (T) and assumed value of P, the partial pressure of each species in vapor (p_i) coexisting with melt can be calculated from the concentrations of volatile components dissolved in the melt. The pressure of vapor-saturation of the melt (P^v_{sat}) is then the unique P at which the sum of all the p_i values matches the total P.

This approach for determining P^v_{sat} for magmas has been widely applied to melt inclusions and matrix glasses, subject to several caveats [e.g., issues related to the fidelity with which glassy melt inclusions preserve the volatile concentrations present at the time of their entrapment: Anderson (1974); Roedder (1979); Anderson and Brown (1993); Portnyagin et al. (2008); Steele-Macinnis et al. (2011); Gaetani et al. (2012); Bucholz et al. (2013); Maclennan (2017); and the possibility of supersaturation of erupting melts with respect to vapor: Fine and Stolper (1986); Dixon et al. (1988); Jendrzejewski et al. (1997); Saal et al. (2002); Soule et al. (2012); Le Voyer et al. (2015, 2019); Aubaud (2022)]. For melt inclusions, this approach has led to many minimum estimates of the P (and therefore depth) of inclusion entrapment (e.g., Anderson et al. 1989; Blundy and Cashman 2008), and such estimates have also been used to "image" magmatic plumbing systems (e.g., Colman et al. 2015; Wanless et al. 2015; Camejo-Harry et al. 2018, 2019; Black and Andrews 2020; Wieser et al. 2021). When applied to sub-aqueously erupted matrix glass, it has been used to estimate eruption depths (e.g., Seaman et al. 2004; Coombs et al. 2006; Lund et al. 2018; Belgrano et al. 2021). This approach also returns the composition of the vapor phase in equilibrium with the melt at P^v_{sat}. This vapor composition can be used to understand whether the melt has undergone closed- or open-system degassing or has been buffered by a large volume of fluid (e.g., Spilliaert et al. 2006; Caricchi et al. 2018; Moretti et al. 2018). The vapor composition can also be used to reconstruct bulk melt inclusion compositions at entrapment if a "shrinkage"

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vapor bubble has developed during cooling, provided that P^v_{sat} and the corresponding vapor composition are calculated at the closure T of bubble formation (e.g., Maclennan 2017; Tucker et al. 2019; Rasmussen et al. 2020).

Most efforts to calculate P^v_{sat} using this approach have assumed that the vapor contains only H_2O and CO_2 and the melt only contains H as hydroxyl ions (OH^-) and/or H_2O molecules (H_2O_{mol}), and C as carbonate ions (CO_3^{2-}) and/or CO_2 molecules ($CO_{2,mol}$) [e.g., VolatileCalc/MIMiC by Newman and Lowenstern (2002) and Rasmussen et al. (2020); MagmaSat by Ghiorso and Gualda (2015); VESIcal by Iacovino et al. (2021); EVo by Liggins et al. (2020, 2022) and Yip et al. (2022); MafiCH by Allison et al. (2022); and various unnamed models such as those presented by Papale et al. (2006), Iacono-Marziano et al. (2012), and Duan (2014)]. These calculations do not generally consider the effects on P^v_{sat} and the corresponding vapor composition of the presence of reduced C-O-H (e.g., H_2, CO, CH_4) and S-bearing species in the vapor and melt. However, oxygen fugacity (f_{O_2}) can play a significant role in the speciation and abundances of C-O-H-S volatile components in coexisting vapor and melt, and S is the third most abundant volatile element in most naturally occurring silicate magmas (e.g., Holloway and Blank 1994). Some studies have considered the importance of S and/or the effects of f_{O_2} during degassing (e.g., Carroll and Webster 1994; Moretti and Papale 2004; Scaillet and Pichavant 2004; Kress et al. 2004; Gaillard et al. 2011; Burgisser et al. 2015; Wetzel et al. 2015; Lesne et al. 2015; Newcombe et al. 2017; Liggins et al. 2020, 2022; Lo et al. 2021; Métrich 2021; Sun and Lee 2022; Wieser et al. 2022; Hughes et al. 2023). However, there has, to our knowledge, not yet been a systematic or quantitative investigation of these effects on calculations of P^v_{sat}.

We are particularly interested in the effects of the so-called "sulfur solubility minimum" [referred to here as the SS^{min}; see Hughes et al. (2023)] on calculated values of P^v_{sat} and vapor composition. The SS^{min} corresponds to a minimum in the dissolved sulfur content of vapor-saturated melt with increasing f_{O_2} at constant P and T in the f_{O_2} range over which the speciation of S dissolved in the melt changes from S^{2-}- to S^{6+}-dominated. This phenomenon for melt + vapor equilibrium has been observed experimentally (e.g., Fincham and Richardson 1954; Katsura and Nagashima 1974; Carroll and Rutherford 1985; Backnaes and Deubener 2011; Lesne et al. 2015; Matjuschkin et al. 2016; Nash et al. 2019) and in various calculations (e.g., Fincham and Richardson 1954; Katsura and Nagashima 1974; Carroll and Rutherford 1985; Moretti et al. 2003; Moretti and Ottonello 2005; Backnaes and Deubener 2011; Baker and Moretti 2011; Lesne et al. 2015; Matjuschkin et al. 2016; Nash et al. 2019; Cicconi et al. 2020; Papale et al. 2022; Hughes et al. 2023). Hughes et al. (2023) used a simplified S-O system (i.e., without C-O-H species) to analyze the factors leading to the SS^{min}. They found the SS^{min} resulted in a maximum in P^v_{sat} with increasing f_{O_2} at constant T and total S content of the melt due to a maximum in p_{SO_2} where the speciation of S in the melt changed from S^{2-}- to S^{6+}-dominated. If this f_{O_2}-dependent maximum in P^v_{sat} extends to C-O-H-S-bearing systems, magma reservoir depths from melt inclusions and eruption pressures from matrix glasses based on calculations of P^v_{sat} that exclude S will be underestimated for magmas with f_{O_2} values in the vicinity of the SS^{min}. Hence, it is important to understand quantitatively the behavior of the SS^{min} in C-O-H-S systems most relevant to natural magmas.

In this paper, we extend the thermodynamic framework of Hughes et al. (2023) to include C-O-H-S vapor and melt species, and we apply it to calculating P^v_{sat}. The model we use is most closely related to those of Gaillard et al. (2011, 2013), Baumgartner et al. (2017), Iacono-Marziano et al. (2017), Liggins et al. (2020, 2022), and Sun and Lee (2022). Our model is written in Python, can be run using Jupyter Notebook, and is available on github (https://github.com/eryhughes/Pvsat). Given this thermodynamic model, we explore predicted trends in P^v_{sat} with varying f_{O_2} for fixed volatile concentrations in the melt. We then explore: (1) how variations in the total amounts of H, C, and S dissolved in the melt influence the calculated value of P^v_{sat}, especially the calculated effects of the predicted buildup of relatively insoluble C-O-H species at low f_{O_2}; and (2) the behavior of the SS^{min} in C- and H-bearing systems and the quantitative impact of this feature on calculated values on P^v_{sat}. Finally, we apply the model to natural melt inclusions and matrix glasses from different tectonic settings on Earth, as well as lunar and martian magmas, to evaluate how calculated values of P^v_{sat} are influenced by these effects.

Calculating the pressure of vapor saturation and the vapor composition

Our goal is to calculate P^v_{sat}, the P at which a silicate melt of a fixed chemical composition (including the concentrations of dissolved volatile components in the melt) can coexist with vapor at a given T and f_{O_2}. At this P, the sum of the p_i values of all species in the vapor (n different species, where each species is denoted with subscript i) in equilibrium with the melt equals the P of the system =

$$P = \sum_{i=1}^{n} p_i \tag{1}$$

and the mole fraction of each species in the vapor (x_i^v) is related to its p_i by:

$$p_i = f_i / \gamma_i^v = x_i^v P \tag{2}$$

where f_i is the fugacity of species i, and γ_i^v is the fugacity coefficient of species i in the vapor (v). Determining P^v_{sat} requires finding the P that is equal to the sum of the p_i values of all molecular species in the vapor, i.e., solving Equation 1. At a higher P than P^v_{sat}, the system is vapor-undersaturated. Hence, these calculations assume the melt is saturated with a vapor—if the melt is not vapor-saturated, the calculated P represents a minimum. At a lower P than P^v_{sat}, melt with the assumed concentrations of volatiles is supersaturated with respect to vapor, and the melt is metastable [or potentially unstable if sufficiently supersaturated; e.g., Allabar and Nowak (2018); Sahagian and Carley (2020); Gardner et al. (2022)]. Given sufficient time, a vapor phase will exsolve from the supersaturated melt, reducing the volatile concentration of the melt such that equilibrium is achieved (i.e., such that Equation 1 is satisfied at the fixed P, T, and bulk composition of the system). Calculating P^v_{sat} for a given melt composition requires choosing: (1) the species in the vapor that will contribute significantly to the total P and the homogenous vapor equilibria that relate them; and (2) solubility functions that provide relationships between the melt composition and the p_i values.

Species and homogeneous equilibria in the vapor

If H_2O and CO_2 are assumed to be the only vapor species, as is typical, Equation 1 becomes:

$$P = \sum_{i=1}^{2} p_i = p_{H_2O} + p_{CO_2}. \tag{3}$$

A review of solubility models used to relate dissolved H_2O and CO_2 concentrations in the melt to p_{H_2O} and p_{CO_2} is provided by Iacovino et al. (2021), which can be used to calculate P^v_{sat} via Equation 3. However, to investigate the effects of f_{O_2} and S, we need to consider a C-O-H-S vapor containing species in addition to those in Equation 3. Although other species could be added, we have considered the following vapor species: O_2, H_2, H_2O, CO, CO_2, S_2, SO_2, H_2S, CH_4, and OCS, such that Equation 1 becomes:

$$P = \sum_{i=1}^{10} p_i = p_{O_2} + p_{H_2} + p_{H_2O} + p_{CO} + p_{CO_2} + p_{S_2} + p_{SO_2} + p_{CH_4} + p_{H_2S} + p_{OCS} \quad (4)$$

The fugacities of these vapor species are related through the six linearly independent homogenous vapor equilibria given in Table 1. Details of our choice of equilibrium constants and fugacity coefficients for vapor species are given in Online Materials[1] Section 3.1.

Solubility functions

When melt and vapor are in equilibrium, there is a function (the "solubility function") that relates the dissolved concentration in the melt of a volatile component to the partial pressure(s) of one (or more) species in the vapor. We use known or likely solubility mechanisms (i.e., chemical reactions between vapor and melt) to constrain the functional forms of the various solubility functions (e.g., Fincham and Richardson 1954; Stolper 1982a; Silver and Stolper 1985; Dixon and Stolper 1995; Gaillard et al. 2011, 2013; Liggins et al. 2020, 2022; Thomas and Wood 2021; Hughes et al. 2023). The key point is that an accurate solubility function allows the calculation of the fugacities of certain vapor species given volatile concentrations in the coexisting melt, from which their p_i values can be calculated using Equation 2.

Based on our chosen vapor species and the assumption of a basaltic melt, we consider the following dissolved C-O-H species in the melt: molecular H_2 ($H_{2,mol}$; e.g., Gaillard et al. 2003; Kadik et al. 2004; Mysen et al. 2011; Hirschmann et al. 2012), hydroxyl ions (OH^-; e.g., Burnham and Davis 1974; Mysen et al. 1980; Stolper 1982a), molecular H_2O (H_2O_{mol}; e.g., Stolper 1982a; Dixon and Stolper 1995; Lesne et al. 2011), molecular CO (CO_{mol}; e.g., Brooker et al. 1999; Armstrong et al. 2015; Yoshioka et al. 2019), carbonate ions (CO_3^{2-}; e.g., Stolper and Holloway 1988; Blank and Brooker 1994; Dixon et al. 1995), and molecular CH_4 ($CH_{4,mol}$; e.g., Ardia et al. 2013; Mysen 2013). For dissolved S, we assume sulfide ions (*S^{2-}, where * refers to sulfide complexes in the melt other than H_2S), sulfate ions (SO_4^{2-}), and molecular H_2S (H_2S_{mol}) are the dominant S-bearing species in the melt (e.g., Fincham and Richardson 1954; Paris et al. 2001; O'Neill and Mavrogenes 2002, 2022; Clemente et al. 2004; Métrich et al. 2009; Wilke et al. 2011; Klimm et al. 2012; Lesne et al. 2015; O'Neill 2021; Boulliung and Wood 2022). Although other S-bearing species have been observed spectroscopically, they are not thought to be significant in natural Fe-bearing melts (e.g., Métrich et al. 2009; Wilke et al. 2011). For each dissolved species in the melt, the corresponding solubility mechanism (heterogeneous melt-vapor equilibrium) is given in Table 2.

The solubility of a particular volatile species can depend on P, T, melt composition, and the presence of other volatile species. To illustrate our approach and the trends that it predicts, we have chosen solubility functions appropriate for a Hawaiian tholeiite (detailed in Table 2). However, other solubility functions can be incorporated into our framework (see Online Materials[1] Section 3.2 for details), and we emphasize that tailored solubility functions for specific melt compositions are required if the goal is detailed petrologic interpretations of particular samples. For $H_{2,mol}$, we use the solubility function of Hirschmann et al. (2012) for basalt but modify it to relate $H_{2,mol}$ in parts per million to f_{H_2} in bar (Eq. 11 in Table 2; see Online Materials[1] Section 3.2.3 for details). For the solubility of H_2O, we use the approximation of Sievert's law; i.e., that f_{H_2O} is proportional to the square of the mole fraction or concentration of the total H dissolved as oxidized species (i.e., combination of OH^- and H_2O_{mol}, expressed as H_2O: H_2O_T; see Eq. 12 in Table 2); this has been shown to be a good approximation up to 6.4 wt% H_2O (e.g., Burnham 1979; Stolper 1982b; Ghiorso and Gualda 2015). We use the data compilation of Allison et al. (2019) to calibrate this solubility function for basaltic melts (see Online Materials[1] Section 3.2.2 for details). For CO_{mol}, we re-parameterize the solubility function of Armstrong et al. (2015) (Eq. 13 in Table 2) using data from Wetzel et al. (2013); Stanley et al. (2014) and Armstrong et al. (2015) (see Online Materials[1] Section 3.2.3 for details). The solubility of CO_3^{2-} is highly dependent on melt composition (e.g., Spera and Bergman 1980; Blank and Brooker 1994; Holloway and Blank 1994; Shishkina et al. 2014; Allison et al. 2019). We use the solubility function of Dixon et al. (1995) (Eq. 14 in Table 2), which is based on data for Hawaiian tholeiites and mid-ocean ridge basalts, although this can be modified for other basaltic compositions (e.g., data from Allison et al. (2019); see Online Materials[1] Section 3.2.1 for details). For $CH_{4,mol}$, we use the solubility function of Ardia et al. (2013) (Eq. 15 in Table 2). We use O'Neill (2021) for the *S^{2-} solubility function and O'Neill and Mavrogenes (2022) for SO_4^{2-} (Eqs. 16 and 17, respectively, in Table 2). We re-parameterize the solubility function for H_2S_{mol} from Clemente et al. (2004) and Lesne et al. (2015) (Eq. 18 in Table 2) using basaltic data from Moune et al. (2009) and Lesne et al. (2015) (see Online Materials[1] Section 3.2.5 for details). We include H_2S_{mol} as a separate species because it is simpler to ensure [compared to its inclusion as another *S^{2-} species; see O'Neill (2021)] that H mass balance is satisfied (i.e., H in the melt is partitioned between H_2O, H_2S, H_2, and CH_4 species) and thus that we do not overestimate the S content of the melt.

In addition to the solubility functions, our thermodynamic model includes homogeneous equilibrium between Fe^{2+} (FeO) and Fe^{3+} ($FeO_{1.5}$) in the melt, where melt Fe^{3+}/Fe^{2+} is related to f_{O_2}, melt composition, P, and T [we use Kress and Carmichael (1991), but other parameterizations can be used]. Finally, we check if the melt is supersaturated with respect to sulfide melt, anhydrite, or graphite; if so, the calculations we present are metastable relative to precipitation of one or more of these phases, and we state this whenever it is the case. The melt is supersaturated with respect to sulfide melt if the *S^{2-} content of the melt is greater than the sulfide content at sulfide saturation [SCSS, assuming the sulfide is pure FeS; O'Neill (2021)]; with respect to anhydrite if the SO_4^{2-} content of the melt is greater

TABLE 1. Homogeneous vapor equilibria considered in our model

Reaction	Equilibrium constant	Eq. no.
$H_2 + 0.5O_2 \rightleftarrows H_2O$	$K_H = \frac{f_{H_2O}}{f_{H_2} f_{O_2}^{0.5}}$	(5)
$CO + 0.5O_2 \rightleftarrows CO_2$	$K_C = \frac{f_{CO_2}}{f_{CO} f_{O_2}^{0.5}}$	(6)
$0.5S_2 + O_2 \rightleftarrows SO_2$	$K_S = \frac{f_{SO_2}}{f_{S_2}^{0.5} f_{O_2}}$	(7)
$CH_4 + 2O_2 \rightleftarrows CO_2 + 2H_2O$	$K_{CH} = \frac{f_{CO_2} f_{H_2O}^2}{f_{CH_4} f_{O_2}^2}$	(8)
$0.5S_2 + H_2O \rightleftarrows H_2S + 0.5O_2$	$K_{HS} = \frac{f_{H_2S} f_{O_2}^{0.5}}{f_{S_2}^{0.5} f_{H_2O}}$	(9)
$2CO_2 + OCS \rightleftarrows 3CO + SO_2$	$K_{SC} = \frac{f_{CO}^3 f_{SO_2}}{f_{CO_2}^2 f_{OCS}}$	(10)

Notes: $K_{X(Y)}$ are equilibrium constants, which depend on T but are independent of P and are given in Online Materials[1] Section 3.1 and f_i is the fugacity of species i.

TABLE 2. Solubility mechanisms and functions considered in our model

Solubility mechanism	Solubility function	Eq. no.
$H_2(v) \rightleftarrows H_{2,mol}(m)$	$f_{H_2} = \frac{w^m_{H_{2,mol}}}{C_{H_{2,mol}}}$	(11)
$H_2O(v) \rightleftarrows H_2O_{mol}(m)$ $H_2O(m) + O^{2-}(m) \rightleftarrows 2OH^-(m)$	$f_{H_2O} \approx \frac{\left(x^m_{H_2O_T}\right)^2}{C_{H_2O_T}}$	(12)
$CO(v) \rightleftarrows CO_{mol}(m)$	$f_{CO} = \frac{w^m_{CO_{mol}}}{C_{CO_{mol}}}$	(13)
$CO_2(v) + O^{2-}(m) \rightleftarrows CO_3^{2-}(m)$	$f_{CO_2} = \frac{x^m_{CO_3^{2-}}}{C_{CO_3^{2-}}}$	(14)
$CH_4(v) \rightleftarrows CH_{4,mol}(m)$	$f_{CH_4} = \frac{w^m_{CH_{4,mol}}}{C_{CH_{4,mol}}}$	(15)
$0.5S_2(v) + O^{2-}(m) \rightleftarrows {}^*S^{2-}(m) + 0.5O_2(v)$	$f_{S_2} = \left(\frac{w^m_{{}^*S^{2-}}}{C_{{}^*S^{2-}}}\right)^2 f_{O_2}$	(16)
$0.5S_2(v) + 1.5O_2(v) + O^{2-}(m) \rightleftarrows SO_4^{2-}(m)$	$f_{S_2} = \left(\frac{w^m_{SO_4^{2-}}}{C_{SO_4^{2-}}}\right)^2 f_{O_2}^{-3}$	(17)
$H_2S(v) \rightleftarrows H_2S_{mol}(m)$	$f_{H_2S} = \frac{w^m_{H_2S_{mol}}}{C_{H_2S_{mol}}}$	(18)

Notes: v = vapor; m = melt; mol = molecular; f_i = fugacity of species i; x_i^m = mole fraction or w_i^m = concentration (depending on the units) of species i in the melt; C_i = solubility of species i, which is the constant of proportionality between the fugacity(ies) and the mole fraction/concentration of species i in the melt; *S^{2-} = sulfide associated with cations in the silicate melt, rather than associated with H, i.e., H_2S_{mol}. Details of solubilities are given in Online Materials[1] Section 3.2.

than the sulfate content at anhydrite saturation (SCAS) (Zajacz and Tsay 2019); and with respect to graphite if f_{CO_2}/f_{O_2} is greater than the equilibrium constant for $C_{graphite} + O_2 = CO_2$ (Holloway et al. 1992) (other parameterizations for saturation of additional phases can be included in our model).

Deriving melt speciation to calculate fugacities and partial pressures

Given P, T, and f_{O_2}, the fugacities of most species in the coexisting vapor can be calculated from concentrations of dissolved volatile species in the melt via Equations 11–18 in Table 2. The homogeneous equilibria in the vapor phase described by Equations 7 and 10 in Table 1 can then be used to calculate the fugacities of the assumed insoluble species SO_2 and OCS, respectively. These fugacities can be converted to partial pressures using Equation 2 and summed together to calculate P^v_{sat} as in Equation 4. However, typically, only the *bulk* concentrations of H, C, and S dissolved in the melt are known, not the concentration of individual H-, C-, and S-bearing *species*. For instance, some analytical techniques used to measure volatile concentrations (e.g., secondary ion mass spectrometry) provide no information on the identity or concentrations of different melt species. However, even if an analytical technique can quantify the concentrations of some species (e.g., Fourier transform infrared spectroscopy for H_2O_{mol} vs. OH^- or $CO_{2,mol}$ vs. CO_3^{2-}), other dissolved species (e.g., H_2, CO, CH_4) are typically not quantifiable at natural abundances. Moreover, the measured speciation could have changed on quenching to a glass (e.g., Stolper 1989; Dingwell and Webb 1990; Morizet et al. 2001; Behrens and Nowak 2010; Konschak and Keppler 2014; Nash et al. 2019).

To deal with this, we solve for the speciation of the melt, subject to the constraints of: the heterogeneous melt-vapor equilibria in Table 2; homogeneous vapor equilibria in Table 1; mass balance given the specified bulk concentrations of H, C, and S in the melt; and the intensive parameters T, f_{O_2}, and P. In this way, the total H, C, and S concentrations measured in the glass are distributed into equilibrium concentrations of the various dissolved species. This is done iteratively, assuming an initial value of P (details of the choice of P are given in the next section). First, all H is assumed to be H_2O_T, all C is CO_3^{2-}, and all S is $*S^{2-}$. Then the concentrations of the other melt species ($H_{2,mol}$, CO_{mol}, $CH_{4,mol}$, H_2S_{mol}, and SO_4^{2-}) are calculated at the given f_{O_2}, T, and P using the equations in Tables 1 and 2. The total amounts of H, C, and S in the melt are then calculated by summing the amounts of these elements in all the melt species. The calculated totals will be higher than the known bulk composition of the melt: e.g., for H, all H dissolved in the melt was initially assumed to be H_2O_T, but after the first iteration, there is additional H in the melt in the form of $H_{2,mol}$, $CH_{4,mol}$, and H_2S_{mol}. Based on the difference between these, new values for the concentrations of H_2O_T, CO_3^{2-}, and $*S^{2-}$ are estimated, and the process is repeated until H, C, and S satisfy mass balance within a defined limit. Note that oxygen is not mass balanced as f_{O_2} is an independent variable in these calculations, and different choices of f_{O_2} lead to different bulk oxygen contents of the melt. Further details are provided in Online Materials[1] Section 4.

These calculations require f_{O_2} as an input. When trying to determine P^v_{sat} for a natural sample, this could be constrained from measurements of Fe^{3+}/Fe_T or S^{6+}/S_T in the glass [e.g., Carroll and Rutherford 1988; Kress and Carmichael 1991, although changes may occur on the quench (Nash et al. 2019)], or by independent oxybarometry. Here, we have chosen f_{O_2} as an independent variable to evaluate the effects of variation in f_{O_2} on P^v_{sat} at constant total dissolved volatile contents. Details are given in Online Materials[1] Section 3.2.6 for how f_{O_2} is chosen and the reference buffers we use in reporting f_{O_2}.

Solving for P^v_{sat} and vapor composition

It is important to note that γ^v_i, solubility, and f_{O_2} all depend on the P chosen for the calculations described above. Therefore, P^v_{sat} must also be calculated iteratively. The initial guess for P (P_0) is chosen to be high (e.g., 40 000 bars). The speciation of the melt is iteratively solved for at the given f_{O_2}, T, P_n, and bulk volatile content of the melt (H, C, and S) as described above. This gives the concentration of specific dissolved species in the melt; f_i (including insoluble vapor species); p_i using Equation 2; and P^v_{sat} using Equation 4. The next guess for P (P_1) is set as the sum of the partial pressures given by the previous guess for P (i.e., $P^v_{sat}[P_0]$), and then the iteration is repeated. After each iteration, the difference between P_n and P^v_{sat} is evaluated: $dP = P_n - P^v_{sat}[P_n]$. If dP after the iteration is less than the specified tolerance (here we use 0.1 bar), P^v_{sat} is taken to have been found. If dP is greater than the specified tolerance, $P_{n+1} = P^v_{sat}[P_n]$, and the calculations of melt speciation and P^v_{sat} are repeated until dP is less than the specified tolerance. Once the calculation has converged, the concentration of all species in the melt and vapor are calculated at P^v_{sat}.

Effects of f_{O_2} and H, C, and S concentrations on vapor-saturation pressures and vapor compositions

To isolate and analyze the effects of f_{O_2} and H, C, and S concentrations on P^v_{sat} and on the composition of the saturating vapor, we use the same base melt composition and T for all calculations in this section. The melt composition is the Hawaiian tholeiite used in Hughes et al. (2023) at $T = 1200$ °C.

P^v_{sat} for a reference case with fixed total volatile concentrations in the melt

In this section, we describe the calculated P^v_{sat} with varying f_{O_2} for a melt containing fixed total dissolved C and H contents (Fig. 1). The total concentration of C is 273 ppm (all parts per million are by weight, corresponding to 1000 ppm of CO_2 if all C were dissolved as CO_2: we refer to this CO_2 equivalent concentration as CO_2-eq) and H is 3357 ppm (corresponding to 3 wt% H_2O-eq). In case 1, all C-O-H species in the melt and vapor are oxidized. Hence, H_2O and CO_2 are assumed to be the only vapor species, and H_2O_T (i.e., the combination of both OH^- and H_2O_{mol}) and CO_3^{2-} are the only volatile-bearing melt species. This is equivalent to most P^v_{sat}-calculators currently in use (e.g., Newman and Lowenstern 2002; Papale et al. 2006; Iacono-Marziano et al. 2012; Duan 2014; Ghiorso and Gualda 2015; Iacovino et al. 2021; Allison et al. 2022). In case 2, both oxidized and reduced C-O-H species are present in the melt and vapor phase. Lastly, we add a fixed amount of S to case 2 (1000 ppm S-eq), leading to case 3, where S-bearing species are present in addition to the C-O-H species included in case 2. Two additional cases are described in Online Materials[1] Section 6.3: case 2′, which is the same as case 2 except reduced C-O-H species are assumed to be completely insoluble in the melt; and case 3′, which is the same as case 3 except H_2S_{mol} is treated as insoluble in the melt.

Case 1: Only oxidized C-O-H species in the melt and vapor

This case will be used as a reference for comparison to the other cases that we consider. For this case, P^v_{sat}, p_i values (p_{CO_2} and p_{H_2O}), melt speciation, and vapor composition ($x^v_{H_2O}$ and $x^v_{CO_2}$) are independent of f_{O_2} (black lines in Figs. 1 and 2). If the solubility functions for CO_3^{2-} and H_2O_T were dependent on the Fe^{3+}/Fe_T of the melt, P^v_{sat}, p_i values, and vapor composition would depend on f_{O_2}. However, melt speciation would still be independent of f_{O_2} because of the assumption that there are no other dissolved C-O-H species. Wieser et al. (2022) found that available models (e.g., Papale et al. 2006; Shishkina et al. 2014; Ghiorso and Gualda 2015) suggest H_2O solubility is relatively insensitive to Fe^{3+}/Fe_T (also observed by Newcombe et al. 2017), whereas CO_2 solubility is sensitive to Fe^{3+}/Fe_T. However, they found the effects are highly uncertain due to a lack of experimental data to calibrate the Fe^{3+}/Fe_T dependence of H_2O and CO_2 solubility. We show a comparison to VolatileCalc (Newman and Lowenstern 2002) in Figure 1 (and other models in Online Materials[1] Section 5, Figs. S7 sand S8), which highlights how much different solubility functions (especially for CO_2) can affect calculations of P^v_{sat}.

Case 2: Oxidized and reduced C-O-H species in the melt and vapor

For case 2, decreasing f_{O_2} results in the fractions of the total H and C dissolved in the melt as reduced species increasing and

as oxidized species decreasing (Fig. 2c), which is reflected in the corresponding p_i values and x_i^v values (Figs. 2a and 2e). H and C behave differently at ΔFMQ < –1, with a large decrease in p_{CO_2} but very little change in p_{H_2O}. This is because the equilibrium constants for homogeneous equilibria 5 and 6 in Table 1 result in p_{CO} beginning to increase significantly with decreasing f_{O_2} (from ~0 to ~1200 bars from ΔFMQ = 0 to –3) at a higher f_{O_2} than that at which p_{H_2} begins to increase. Hence, the amount of CO_3^{2-} in the melt decreases substantially over the f_{O_2} range covered in the figure, whereas there is less change in the amount of H_2O_T (i.e., compare the solid blue and red curves in Fig. 2c).

For ΔFMQ ≳ 0, P^v_{sat} is approximately equal to that assuming the vapor only contains CO_2 and H_2O (Fig. 1) because the p_i values and x_i^v values of H_2, CO, and CH_4 are negligible in the vapor (red and blue curves in Figs. 2a and 2e). With decreasing f_{O_2} below ΔFMQ ≈ 0, the p_i values of these reduced vapor species increase, while those of H_2O and CO_2 decrease, but the increases for the reduced species are larger (Fig. 2a). Hence, there is a significant increase in P^v_{sat} with decreasing f_{O_2} when the reduced C-O-H species are included (Fig. 1). This is due to the buildup of reduced C-O-H species in the system that are less soluble than oxidized C-O-H species (e.g., Scaillet and Pichavant 2004; Wetzel et al. 2015; Newcombe et al. 2017). If the reduced C-O-H species are assumed to be completely insoluble in the melt, the increase in P^v_{sat} is even greater (see Online Materials[1] Section 6.3.1). For the conditions shown in Figure 2, graphite is not stable.

Case 3: S-bearing species in the melt and vapor

The variations with f_{O_2} for the C- and H-bearing vapor and melt species for the S-bearing case 3 are similar to those for the S-free case 2 (i.e., compare the blue and red curves Fig. 2). This reflects: (1) the relatively low concentration of S in the melt (i.e., H_2O-eq = 3 wt% vs. S-eq = 0.1 wt%); and (2) the relatively high solubility of S in the melt compared to CO_2 such that the vapor has a high C/S ratio relative to the melt. Hence, the p_i values of S-bearing species are small and a small fraction of P^v_{sat} under all conditions shown in Figure 2b. Another factor is the assumed insolubility of OCS species in the melt for this case, which means dissolved S has a negligible effect on the concentrations of C-bearing species in the melt. For the conditions shown in Figure 2, graphite, sulfide, and anhydrite are not stable.

Sulfur is dissolved in the melt nearly entirely as S^{2-} at low f_{O_2} (ΔFMQ ≲ 0) and SO_4^{2-} at high f_{O_2} (ΔFMQ ≳ 2) (Fig. 2d; e.g., Fincham and Richardson 1954; Moretti and Ottonello 2005, 2003; Moretti and Papale 2004; Gaillard and Scaillet 2009, 2014; Baker and Moretti 2011; Gaillard et al. 2011, 2013, 2015; Baumgartner et al. 2017; Moretti 2021). Additionally, at low f_{O_2} (ΔFMQ ≲ 0), H_2S_{mol} and $*S^{2-}$ are present in roughly equal molar concentrations in the melt (Fig. 2d). By combining Equations 9, 16, and 18 in Tables 1 and 2, we obtain the following expression for the ratio of S^{2-} dissolved in the melt as H_2S_{mol} to that dissolved as $*S^{2-}$:

$$\frac{x^m_{H_2S_{mol}}}{x^m_{*S^{2-}}} = \frac{K_{HS}C_{H_2S_{mol}}}{C_{*S^{2-}}} f_{H_2O} \tag{19}$$

Therefore, $H_2S_{mol}/*S^{2-}$ increases linearly with increasing f_{H_2O} [or equivalently with $(x^m_{H_2O_T})^2$ based on Eq. 12 in Table 2] if the melt composition is otherwise fixed (i.e., at given f_{H_2O}, P, and T,

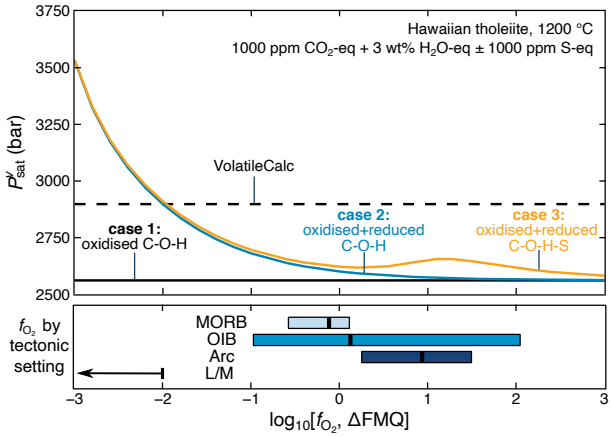

Figure 1. Calculated values of P^v_{sat} vs. $\log_{10}(f_{O_2})$ for a Hawaiian tholeiitic melt at 1200 °C with H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq = 0 ppm [cases 1 and 2, black horizontal lines and blue curves, respectively, from this study; and the horizontal black-dashed line is similar to case 1 but calculated using VolatileCalc (Newman and Lowenstern 2002)] or S-eq = 1000 ppm (case 3, yellow curve). For these conditions, none of the melt compositions are supersaturated with respect to graphite, sulfide, or anhydrite. The f_{O_2} ranges for different tectonic settings from Cottrell et al. (2021) are shown by horizonal blue bars below the main figure (the extent of each bar shows the range and the black vertical line shows the mode) for mid-ocean ridge (MORB: light blue), ocean island (OIB: intermediate blue), and arc (Arc: dark blue) basalts based on XANES measurements of Fe^{3+}/Fe_T of glass. The upper limit of f_{O_2} for lunar and martian magmas (L/M) is shown by a capped black horizontal arrow (Wadhwa 2008; Gaillard et al. 2021). (Color online.)

the $H_2S_{mol}/*S^{2-}$ ratio depends only on the ratio of the solubilities in Eq. 19). The ratio is independent of f_{S_2} and f_{O_2} (except through the dependence of f_{H_2O} on f_{O_2} and f_{H_2} through reaction 5 in Table 1). The relationship between S^{6+}/S_T^{2-} and f_{O_2} (where S_T^{2-} is the total S dissolved as sulfide, given by $*S^{2-} + H_2S$) can be derived by combining Equations 17 (Table 2) and 19:

$$\frac{x^m_{SO_4^{2-}}}{x^m_{S_T}} = \frac{x^m_{SO_4^{2-}}}{x^m_{H_2S_{mol}} + x^m_{*S^{2-}}} = \frac{C_{SO_4^{2-}}}{C_{H_2S_{mol}} K_{SH} f_{H_2O} C_{*S^{2-}}} \left(f_{O_2}\right)^2$$
$$\approx \left[\frac{C_{SO_4^{2-}}}{K_{HS} C_{H_2S_{mol}} \frac{\left(x^m_{H_2O_T}\right)^2}{C_{H_2O_T}} + C_{*S^{2-}}}\right] \left(f_{O_2}\right)^2 \tag{20}$$

Hence, SO_4^{2-}/S_T^{2-} depends on $(f_{O_2})^2$, as is the case for $SO_4^{2-}/*S^{2-}$ (e.g., Wallace and Carmichael 1994; Matthews et al. 1999; Moretti and Ottonello 2005; Métrich et al. 2009; Jugo et al. 2010; Moretti 2021; Hughes et al. 2023), but with different slopes.

At all values of f_{O_2} covered by our modeling, p_{S_2} contributes negligibly to the sum of the p_i values of the S-bearing species (Fig. 3). At f_{O_2} ≲ ΔFMQ ~ +0.4, H_2S is predicted to become the dominant S-bearing vapor species, followed by OCS (Fig. 3). The decreases in p_{H_2S} and p_{OCS} with increasing f_{O_2} seen in Figure 3 between +0.40 ≲ ΔFMQ ≲ +1.5 results from the proportionality of both f_{H_2S} (Eq. 9; Table 1) and f_{OCS} (Eqs. 6, 7, and 9; Table 1) to $(f_{O_2})^{-0.5}$. As p_{H_2S} and p_{OCS} decrease, p_{SO_2} increases with increasing f_{O_2}, with p_{SO_2} surpassing p_{H_2S} at ~ΔFMQ+0.4 (Fig. 3). For this melt

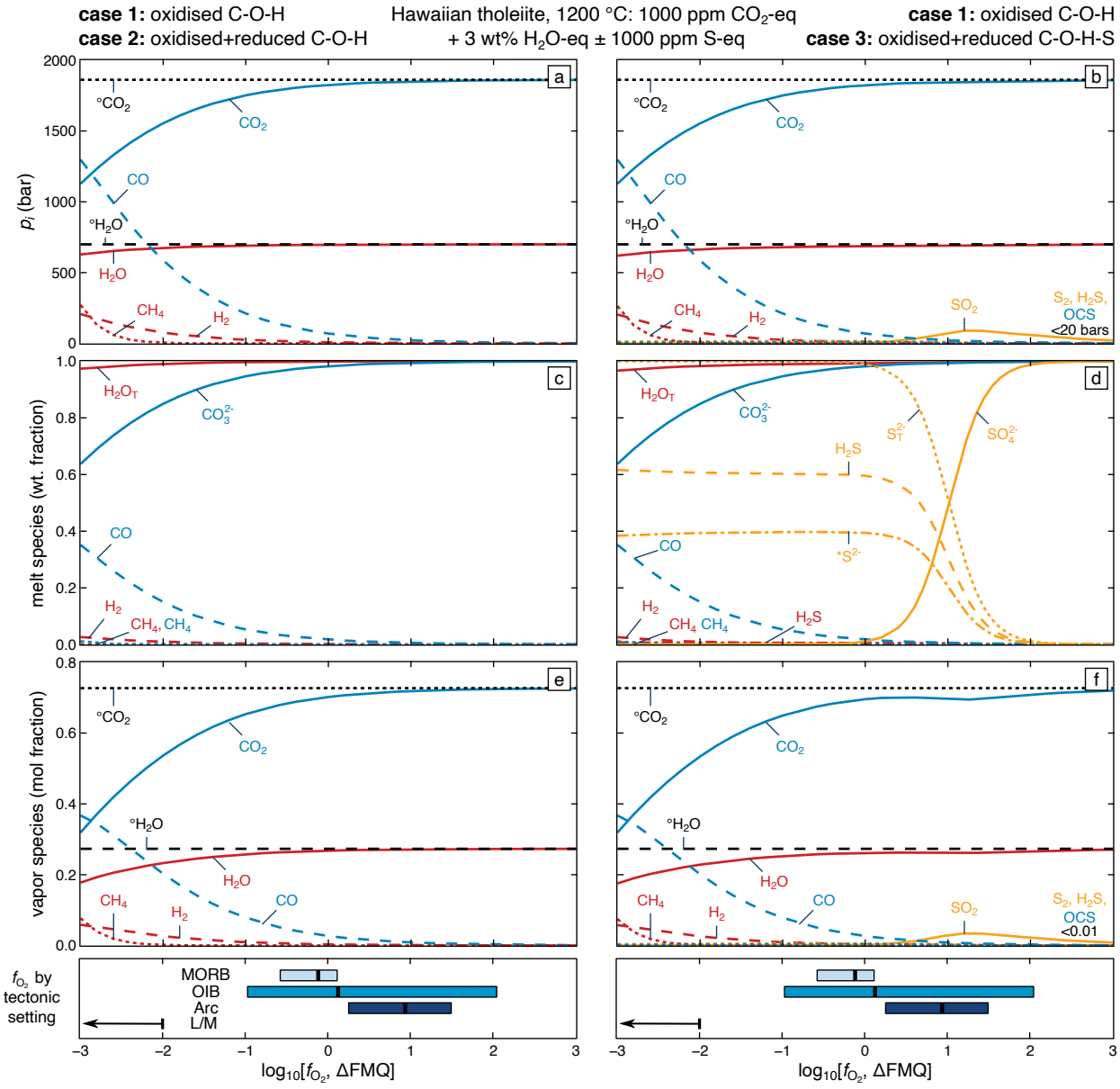

FIGURE 2. Calculated values of p_i (top row), species fraction in the melt (middle row), and x_i^v (bottom row) vs. $\log_{10}(f_{O_2})$ for cases 1–3 discussed in the text. All calculations are for a Hawaiian tholeiitic melt at 1200 °C with H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq = 0 ppm (case 1 shown in both columns and case 2 shown in column 1) or S-eq = 1000 ppm (case 3 shown in column 2). The red curves in both columns are for H-bearing species (including CH_4, but not H_2S); blue curves for C-bearing species (including OCS, but not CH_4); and yellow curves in column 2 are for S-bearing species (including H_2S, but not OCS). Results for case 1 (indicated by a superscript circle, °) are shown for reference in **a**–**b** and **e**–**f**: H_2O is a dashed horizontal black line and CO_2 is a dotted horizontal black line. For melt speciation in **c**–**d**, the blue curves show the fractions of C in different C-bearing species as a fraction total C in the melt (CO_2 = solid, CO = dash, CH_4 = dot), and these fractions sum to one. The red curves show the fractions of H in different H-bearing species as a fraction total H in the melt (H_2O_T = solid, H_2 = dash, CH_4 = dot, H_2S = dot-dash), which also sum to one. The yellow curves show the fractions of S in different S-bearing species as a fraction total S in the melt (SO_4^{2-} = solid, $*S^{2-}$ = dash, H_2S = dot-dash, $S_T^{2-} = *S^{2-} + H_2S$ = dot). For the S-bearing melt species shown in **d**, both $[SO_4^{2-} + *S^{2-} + H_2S]$ and $[SO_4^{2-} + S_T^{2-}]$ equal one. For the vapor speciation in **e**–**f**, the sum of all colored curves is one at any given f_{O_2}, and likewise for the horizontal black reference lines representing case 1. The values of p_i and x_i^v for S_2, H_2S, and OCS in **b** and **f** are always near zero (i.e., close to the *x*-axis) on this scale (see Fig. 3 for a zoomed in version). For these conditions, none of the melt compositions are supersaturated with respect to graphite, sulfide, or anhydrite. The f_{O_2} ranges for different settings are as described in Figure 1. (Color online.)

composition, there is a maximum in p_{SO_2} at ΔFMQ+1.28, where both S_T^{2-} and SO_4^{2-} species are dissolved in the melt in subequal amounts (Figs. 3 and 4; there is also a much smaller maximum in p_{S_2} at ΔFMQ+0.77). When S_T^{2-} is the dominant form of S dissolved in the melt, p_{S_2} and p_{SO_2} are, respectively, proportional to $(f_{O_2})^1$ and $(f_{O_2})^{1.5}$ (Eqs. 6, 7, and 16 in Tables 1 and 2). In contrast, when SO_4^{2-} dominates, p_{S_2} and p_{SO_2} are, respectively, proportional to $(f_{O_2})^{-3}$ and $(f_{O_2})^{-0.5}$ (Eqs. 6, 7, and 17 in Table 2). As discussed in Hughes et al. (2023), this change in the sign of the dependence of the derivative of p_{SO_2} and p_{S_2} with respect to f_{O_2} results in a maximum in p_{SO_2} and p_{S_2} in a C- and H-free system. The same effect produces the maximum in P^v_{sat} at ΔFMQ+1.19 demonstrating the SS^{min} is a general feature of S-bearing basaltic magmas at relatively oxidizing conditions (Fig. 3).

Even though S-bearing species are minor components of the vapor ($x^v_i < 0.03$, where $i = S_2, SO_2, H_2S$, OCS; Fig. 2f), their presence in case 3 influences P^v_{sat}. At all the f_{O_2} values shown, P^v_{sat} for case 3 is greater than case 2 (compare the yellow and blue curves in Fig. 1). At low f_{O_2} (ΔFMQ <0), this is due to the addition of H_2S and OCS to the vapor for case 3 (by up to ~20 bars at ΔFMQ = 0), which are not present in case 2. Additionally, the changing behavior of S-bearing species in the melt and vapor with f_{O_2} (mainly H_2S and SO_2) results in the SS^{min} seen in Figure 1. The overall behavior of case 3 shown in Figure 3 is similar to that calculated for the S-O (i.e., H- and C-absent, referred to as case 4) system presented in Hughes et al. (2023). However, the magnitude and f_{O_2}-position of the maximum in P^v_{sat} are different, reflecting the influence of C and H on the system (more details in Online Materials[1] Section 6.2).

Lesne et al. (2015) calculated $p_{H_2S} + p_{SO_2}$ for vapor coexisting with a basaltic melt in a C-free system (i.e., the vapor was assumed to contain only S-O-H species) for varying f_{O_2} at constant dissolved S-eq, dissolved H_2O-eq, T, and P. They predicted that ignoring

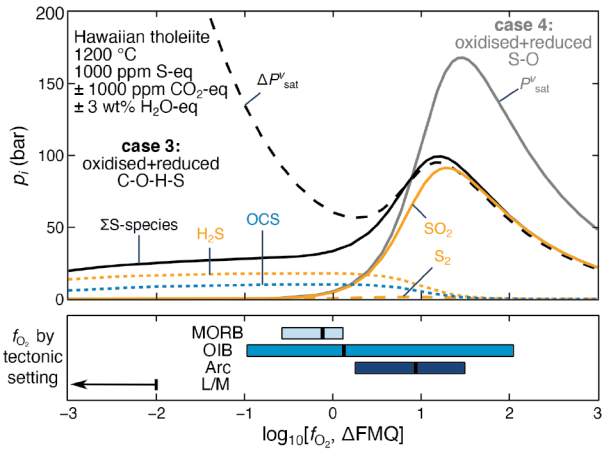

Figure 3. Variations with $\log_{10}(f_{O_2})$ for a Hawaiian tholeiitic melt at 1200 °C with H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq 1000 ppm (case 3) of the partial pressures of individual S-bearing vapor species (S_2 = dash, SO_2 = solid, and H_2S = dot in yellow; OCS = blue dash), their sum [ΣS-species = $p_{S_2} + p_{SO_2} + p_{H_2S} + p_{OCS}$ in solid black), ΔP^v_{sat} (dash black), and P^v_{sat}[case 4] ([S-O] = gray; i.e., $p_{S_2} + p_{SO_2} + p_{O_2}$ for S-eq = 1000 ppm, CO_2-eq = 0 ppm, and H_2O-eq = 0 wt% (Hughes et al. 2023)]. For these conditions, none of the melt compositions are supersaturated with respect to graphite, sulfide, or anhydrite. The f_{O_2} ranges for different settings are as described in Figure 1. (Color online.)

S-bearing vapor species could lead to underestimates of P^v_{sat} of 10^2–10^3 bars at ~2000 bars. However, in contrast to our results, their model produced a local minimum in P^v_{sat}, stating:

> "It is only within the minimum of sulfur solubility, i.e., around NNO±0.5 [~ΔFMQ+0.2 to +1.2], that ignoring partial pressure of S-bearing species does not greatly affect gas saturation calculations."

The behavior predicted by Lesne et al. (2015) assumes that the only S-bearing species in the melt are H_2S_{mol} and $SO_{2,mol}$. However, there is evidence for other dissolved S^{2-} species under reducing conditions in addition to H_2S_{mol} in natural silicate melt compositions (e.g., FeS, CaS; O'Neill 2021). Additionally, there is no evidence for $SO_{2,mol}$ in natural silicate melt compositions under oxidizing conditions, whereas SO_4^{2-} has been observed spectroscopically (e.g., Wilke et al. 2011). Therefore, we infer that SS^{min} will result in a local maximum in P^v_{sat} for silicate melts in nature rather than a local minimum. However, if there are systems where the dominant oxidized S-bearing species in the melt is $SO_{2,mol}$ and H_2S_{mol} dominates over other dissolved sulfide species at reduced conditions, a minimum in P^v_{sat} might occur.

Varying the concentration of volatiles

We now investigate how variations in the concentrations of volatiles (i.e., bulk concentrations of C, H, and S in the melt) affect the relationship between P^v_{sat} and f_{O_2}. This calculation uses the full C-O-H-S system with the same melt and vapor species as in case 3, but we allow the dissolved total volatile contents to vary between 0–5 wt% H_2O-eq, 0–2000 ppm CO_2-eq, and 0–2000 ppm S-eq.

We first calculate P^v_{sat} for a fully-oxidized, S-free reference system comparable to case 1 between 0–5 wt% H_2O-eq and 0–2000 ppm CO_2-eq, shown as isobars in Figure 4a. The differences between this reference case and the full C-O-H-S system are contoured in Figures 4b–4d, where the contours are for constant values of $\Delta P^v_{sat} = P^v_{sat}$[C-O-H-S] – P^v_{sat}[oxidized C-O-H reference state]. Case 3 (3 wt% H_2O-eq, 1000 ppm CO_2-eq, and 1000 ppm S-eq) is shown by the dashed gray horizontal lines in Figures 4b–4d. Figure 4b has variable S (0–2000 ppm S-eq) at constant H (3 wt% H_2O-eq) and C (1000 ppm CO_2-eq); Figure 4c has variable H (0–5 wt% H_2O-eq) at constant S (1000 ppm S-eq) and C (1000 ppm CO_2-eq); and Figure 4d has variable C (0–2000 ppm CO_2-eq) at constant H (3 wt% H_2O-eq) and S (1000 ppm S-eq). In the following two subsections, we emphasize two features of Figure 4: (1) The increase in P^v_{sat} with decreasing f_{O_2} at low f_{O_2} (ΔFMQ $\lesssim$ 0) due to the buildup of relatively insoluble reduced C-O-H species in the vapor; and (2) the local maximum in P^v_{sat} at intermediate f_{O_2} (ΔFMQ ≈ +0.5 to +1.5 for this melt composition), which is a manifestation of the SS^{min}.

Buildup of relatively insoluble reduced C-O-H species in the vapor

At any given y-value on Figures 4b–4d (i.e., a horizontal slice), ΔP^v_{sat} increases with decreasing f_{O_2} for ΔFMQ $\lesssim$ 0. This is due to the buildup of reduced C-O-H species in the vapor because these species are less soluble in the melt than their oxidized counterpart species (e.g., Scaillet and Pichavant (2004); Wetzel et al. (2015); Newcombe et al. (2017); discussed for case 2 above). The contours of ΔP^v_{sat} also get closer together with decreasing

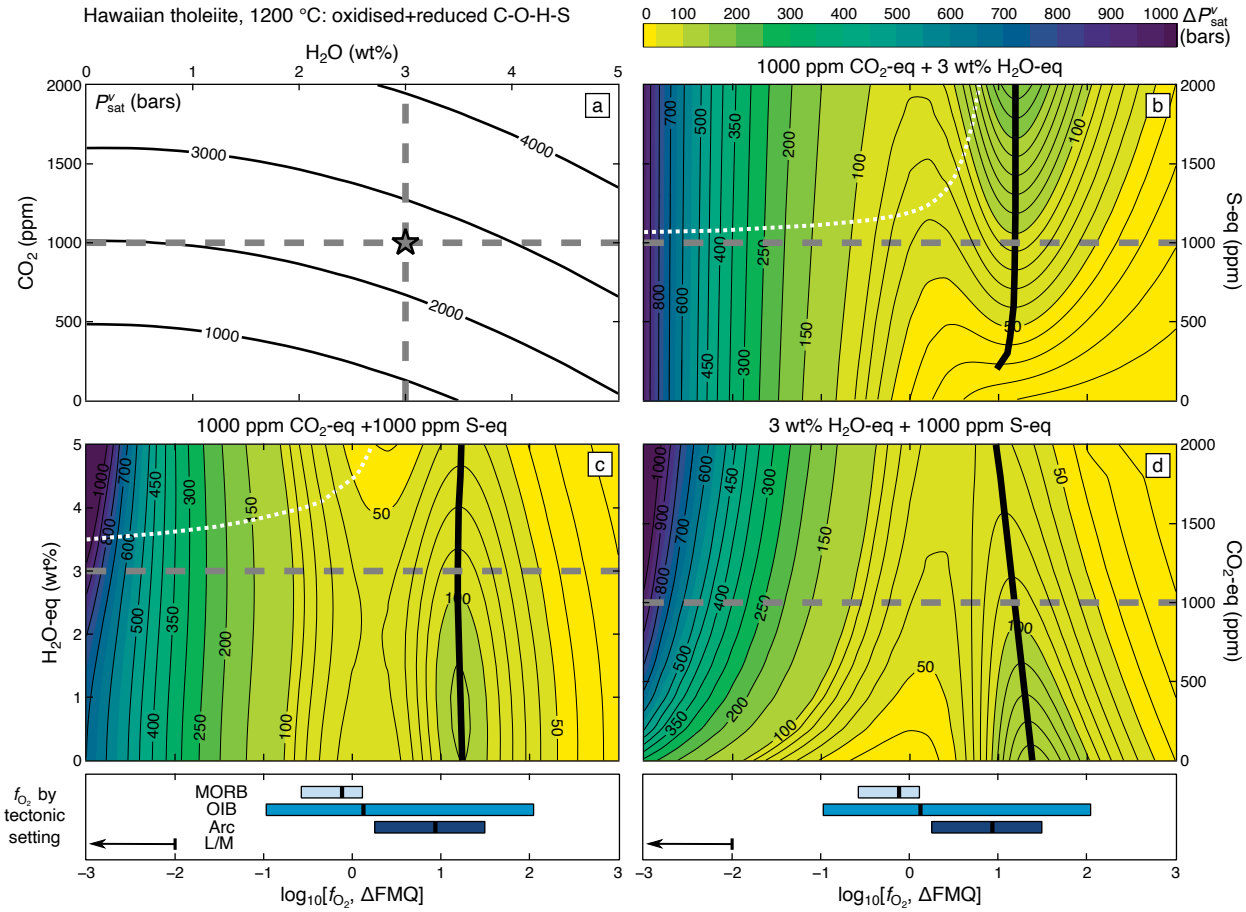

Figure 4. Effects of varying volatile concentrations (S, H, and C) and f_{O_2} on P^v_{sat} for a Hawaiian tholeiitic melt at 1200 °C. (**a**) CO_2 against H_2O dissolved in vapor-saturated melt contoured for P^v_{sat} (bars; black solid curves) calculated assuming case 1. The gray dashed lines in **a** are the H_2O-CO_2 concentrations shown in **b**–**d** and the gray star is case 1. (**b**–**d**) Contours of ΔP^v_{sat} (bars, difference between case 3 [C-O-H-S] and case 1 [oxidized C-O-H]): (**b**) H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq = 0–2000 ppm; (**c**) H_2O-eq = 0–5 wt%, CO_2-eq = 1000 ppm, and S-eq = 1000 ppm; and (**d**) H_2O-eq = 3 wt%, CO_2-eq = 0–2000 ppm, and S-eq = 1000 ppm. The gray dashed lines in **b**–**d** correspond to case 3 and illustrated in Figures 1, 2b, 2d, 2f, and 3 (H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq = 1000 ppm), and the black solid curves are the trace of the maximum in P^v_{sat}. For these conditions, none of the melt compositions are supersaturated with respect to graphite or anhydrite, but are supersaturated with respect to sulfide above the white dotted curve. The f_{O_2} ranges for different settings are as described in Figure 1. (Color online.)

$\log_{10}(f_{O_2})$, reflecting the concave up-shapes of the curves shown in Figure 1. However, the magnitude of the predicted effects—and even their signs—could be changed if future experiments were to demonstrate significantly lower or higher solubilities of these reduced species.

Contours of ΔP^v_{sat} with constant H and C contents but variable S contents are near-vertical; i.e., variations in the S content of the melt have only small effects on ΔP^v_{sat} (Fig. 4b). This is because S-bearing species represent such small proportions of the vapor at low f_{O_2} (e.g., <30 bars with 1000 ppm S-eq for ΔFMQ $\lesssim$ 0; black solid curve in Fig. 3). Although the S content of the melt is changing along each contour in Figure 4b, neither the direct impact of these changes through variations of the p_i values of S-bearing species nor their indirect effects on the p_i values of the C-O-H species result in significant effects on P^v_{sat}.

Contours of ΔP^v_{sat} with constant S and C contents but variable H contents are also near-vertical (Fig. 4c), whereas for constant H and S but variable C, they are steeply, positively sloped, and concave up (Fig. 4d). Therefore, increasing the C content of the melt has a greater effect on increasing ΔP^v_{sat} with decreasing f_{O_2} than increasing the H content of the melt, even though the absolute concentration of C is far smaller. This reflects the increase in the relatively insoluble CO-species in the vapor with decreasing f_{O_2} by conversion from CO_2 to CO (reaction 6, Table 1), which occurs at a higher f_{O_2} than the increase in the H_2 species by conversion from H_2O with decreasing f_{O_2} (reaction 5, Table 1) (described for case 2 above).

Graphite and anhydrite are not stable under the conditions shown in Figure 4. However, sulfide is stable above the white dotted curves in Figures 4b and 4c (but nowhere in Fig. 4d because the S and H content of the melt are too low), and hence, the calculations are metastable in this region [see Hughes et al. (2023) for explanations of the shape of the sulfide-stable curve]. This shows that sulfide is stabilized at low f_{O_2} (ΔFMQ < 0), high

dissolved S (>1000 ppm S), and high dissolved H_2O (>3.5 wt% H_2O-eq because of the effect of dilution on the SCSS).

The sulfur solubility minimum in C- and H-bearing melts

A near-vertical "ridge" defining a maximum in ΔP^v_{sat} at +0.5 < ΔFMQ < +1.5 is visible in Figures 4b–4d. The trace of this ridge with varying volatile contents is shown in each panel as a black curve. For ease of comparison, these three traces are superimposed in Figure 5a, where volatile concentrations are plotted vs. the height of the ridge (i.e., ΔP^v_{sat}), and in Figure 5b, where volatile concentrations along the "ridge" are plotted vs. the f_{O_2}-position of the ridge.

The ΔP^v_{sat} maximum in Figures 4b–4d is the signature of the SS^{min}, and it is also closely associated with maxima in the sum of the p_i values of S-bearing vapor species [especially p_{SO_2}, Fig. 3, and Hughes et al. (2023)]. Hence, the ridge in calculated ΔP^v_{sat} disappears in the S-free systems (i.e., near the *x*-axis in Fig. 4b) and increases in height with increasing total dissolved S in the melt coexisting with vapor (e.g., with increasing S in Fig. 4b). The interactions among C-, H-, and S-bearing species in the melt and vapor result in the systems described here being more complex than the simple cases considered by Hughes et al. (2023) (described for case 3 above). Nevertheless, our modeling predicts that this feature persists robustly in basaltic systems saturated with C-O-H-S vapor.

The f_{O_2} value of the maximum in ΔP^v_{sat} varies little (+0.97 < ΔFMQ < +1.38) over the ranges in S-eq, H_2O-eq, and CO_2-eq explored here (Fig. 5b). In contrast, the height of the ΔP^v_{sat} maximum does depend on the volatile contents of the melt (Fig. 5a). The maximum in ΔP^v_{sat} reflects competing reactions involving dissolved S (described for case 3 above), e.g., Hughes et al. 2023). Hence, its magnitude increases with increasing S-eq because p_{SO_2} is proportional to the total sulfur content of the melt (yellow line in Fig. 5a; derivation in Online Materials[1] Section 6.1). However, $\Delta P^v_{sat} \neq 0$ at S-eq = 0 ppm, reflecting the presence of small but non-negligible quantities of reduced C-O-H species in the vapor such that $\Delta P^v_{sat} > 0$. The buildup of relatively insoluble C-O-H species (especially CO) contributes sufficiently to the ΔP^v_{sat} value that it begins to interfere with and obscures the SS^{min} at low S such that the maximum in ΔP^v_{sat} disappears before S-eq reaches zero (e.g., yellow curves in Fig. 5 terminate at S-eq > 0). Increasing CO_2-eq in the melt causes the magnitude of the ΔP^v_{sat} maximum to decrease. The magnitude of the ΔP^v_{sat} maximum first increases then decreases with increasing H_2O-eq, highlighting the complex interplay between H-, C-, and S-bearing melt and vapor species.

Implications

Figure 4 emphasizes that calculations of the magnitude and exact position of the SS^{min} and their impact on calculations of P^v_{sat} will depend on precise knowledge of the volatile solubility functions, through which the complex interplay discussed in the previous section compositions must ultimately be modeled. Complete and accurate parameterizations of the solubilities in Table 2 in terms of the major element composition of the melt, f_{O_2}, and the concentrations of each volatile component are needed for accurate calculations of P^v_{sat}. It is important to be aware of these uncertainties to avoid over-interpretation of the calculated absolute values of P^v_{sat}. For example, uncertainty in the parameters used in the model (e.g., solubilities and their parameterization, vapor fugacity coefficients) and analytical errors on the melt composition (major, minor, and volatile species, as well as iron oxidation state) can result in significant uncertainties in the accuracy of calculated values of P^v_{sat} (see Online Materials[1] Sections 7.6 and 7.7). The largest source of uncertainty in calculations of P^v_{sat} are the carbonate solubilities and their significant dependence on major element composition of the melt and errors on measurements of CO_2-eq (and H_2O-eq) concentration, which can dwarf variations due to f_{O_2} and S (Online Materials[1] Fig. S17). Despite these uncertainties, we are confident that our modeling captures the overall trends in the effects of S and f_{O_2} on calculations of P^v_{sat} and ΔP^v_{sat} for basaltic melt compositions (see Online Materials[1] Section 7.2). We describe these potential limitations of our model in detail in Online Materials[1] Section 7.

As emphasized in the Introduction, values of P^v_{sat} are routinely and widely calculated based on the volatile contents of glasses to provide minimum estimates of the *P* at which the volatile contents were set (e.g., Anderson et al. 1989; Blundy and Cashman 2008; Colman et al. 2015; Wanless et al. 2015; Camejo-Harry et al. 2018,

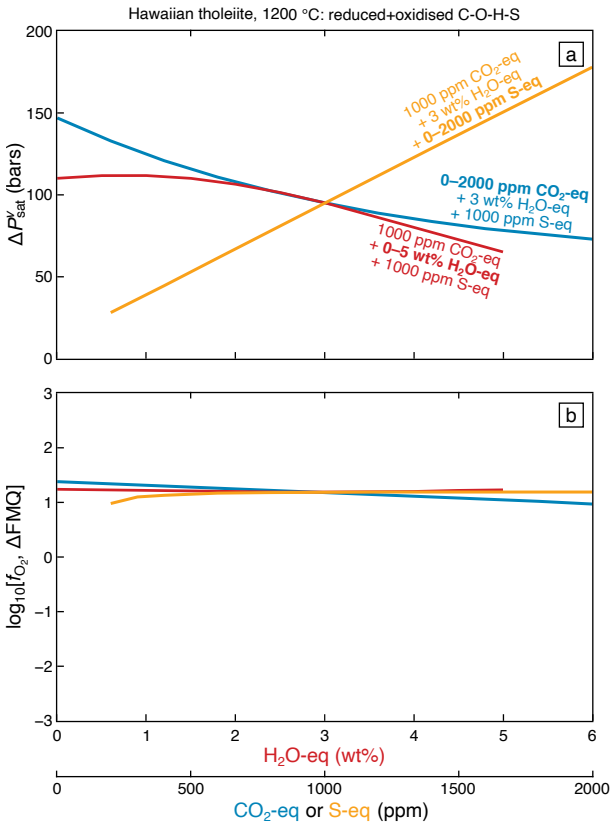

Figure 5. The (**a**) magnitude and (**b**) position of the maximum in P^v_{sat} for a Hawaiian tholeiitic melt at 1200 °C using case 3 [C-O-H-S] containing: H_2O-eq = 3 wt%, CO_2-eq = 1000 ppm, and S-eq = 0–2000 ppm (yellow); H_2O-eq = 0–5 wt%, CO_2-eq = 1000 ppm, and S-eq = 1000 ppm (red); and H_2O-eq = 3 wt%, CO_2-eq = 0–2000 ppm, and S-eq = 1000 ppm (blue). The *x*-axis is the concentration of volatile species that is changing for each curve (e.g., for the yellow curves it is S-eq). Although rotated, the curves in **b** are the same as the black curves in Figures 4b–4d. (Color online.)

2019; Black and Andrews 2020; Wieser et al. 2021). With advances in microanalysis, melt inclusion and matrix glass studies are increasingly co-determining the concentrations of a complete suite of volatiles in the samples (H, C, S, Cl, F, etc.) and Fe^{3+}/Fe_T and/or S^{6+}/S_T to constrain f_{O_2} (e.g., Rose-Koga et al. 2021). However, few calculations of P^v_{sat} include the effects of dissolved S or f_{O_2} (e.g., Carroll and Webster 1994; Newman and Lowenstern 2002; Lesne et al. 2015; Iacovino et al. 2021; Yip et al. 2022).

To investigate the magnitude of these effects in nature, we collated literature data on glassy melt inclusions and matrix glasses where CO_2, H_2O, S, and Fe^{3+}/Fe_T have been measured for the same glass sample. We have restricted our compilation to low-alkali basaltic glass compositions similar to the Hawaiian tholeiite we have adopted as a reference as the carbonate solubility is highly dependent on melt composition and is fixed in these calculations (e.g., Online Materials[1] Fig. S15). However, to examine magmas formed at the low-f_{O_2} extreme of our calculations, we include lunar and martian samples. These have much higher FeO than a Hawaiian tholeiite, and hence, the carbonate solubility function we use here may not be appropriate. For each terrestrial glass sample included in our compilation, we calculate P^v_{sat} at $T = 1200$ °C using the known f_{O_2} based on measured Fe^{3+}/Fe_T ratios and measured melt composition (which can influence $C_{*S^{2-}}$ and $C_{SO_4^{2-}}$ in our model, but not other solubilities). For lunar and martian glasses, we calculate P^v_{sat} using a range of f_{O_2} appropriate for these magmas [$-5 < \Delta FMQ < -1$; e.g., Shearer et al. (2006), Gaillard et al. (2021)]. As well as P^v_{sat}, we also calculate the vapor composition in equilibrium with the measured melt composition at P^v_{sat}: we do not discuss the vapor composition in the main text, but this can be found in Online Materials[1] Section 8. Calculated vapor compositions could be compared to natural volcanic gas samples and fluid inclusions or used to reconstruct bulk melt inclusion compositions when a bubble is present.

Most modern terrestrial magmas have f_{O_2} levels corresponding to $-1 \lesssim \Delta FMQ \lesssim +2$ (e.g., Cottrell et al. 2021) and contain significant S (typically 900–1200 ppm S for MORBs, 1100–6000 ppm S for intraplate hotspots, and 1200–7000 ppm S for arcs; Wallace and Edmonds 2011; Gaillard et al. 2021). Hence, the SS^{min}, which for our calculations is between $+1 \lesssim \Delta FMQ \lesssim +2$ (Fig. 1) could influence calculated P^v_{sat} values for hotspot and arc magmas that have f_{O_2} values overlapping the calculated range of the SS^{min}. The buildup of relatively insoluble reduced C-O-H species in the vapor described previously (Fig. 1) is unlikely to be important for typical terrestrial magmas but is likely to be important for extra-terrestrial settings. Although we also discuss values of absolute P^v_{sat}, we focus on ΔP^v_{sat}, the difference between P^v_{sat} for full C-O-H-S-bearing system and the P^v_{sat} value for the oxidized C-O-H reference (i.e., case 1, which has been used widely in existing calculations of P^v_{sat}).

Mid-ocean ridge basalts

Figures 6a–6c show measured volatile contents and calculated P^v_{sat} and ΔP^v_{sat} for MORB submarine matrix glasses. These MORBs have an $f_{O_2} \leq \Delta FMQ+0.4$, so their calculated values of P^v_{sat} are relatively insensitive to f_{O_2} and S at the S contents observed ($\Delta P^v_{sat} \leq 33$ bars, Fig. 6c). Thus, calculations of P^v_{sat} for MORB that do not include the effects of f_{O_2} and S would only underestimate P^v_{sat} by $\lesssim$30 bars (Fig. 6c). Reykjanes Ridge glasses have low CO_2 and H_2O concentrations due to degassing to shallow depths [$\leq$100 ppm CO_2-eq and $\leq$0.3 wt% H_2O-eq; Fig. 6a; e.g., Nichols et al. (2002)], but they have not degassed significant S [~1200 ppm S-eq, Fig. 6b; e.g., Nichols et al. (2002), Shorttle et al. (2015)] and their f_{O_2} is roughly where the SS^{min} can start to affect P^v_{sat} (Fig. 6b). The combination of relatively high f_{O_2} and S-eq and low H_2O-eq and CO_2-eq leads to large values of $\Delta P^v_{sat}/P^v_{sat}$ [i.e., 10–40% based on the lines of constant $100\cdot(\Delta P^v_{sat}/\Delta P^v_{sat})$ shown in Fig. 6c]. Thus, including the effects of S and f_{O_2} in calculations of P^v_{sat} results in only small absolute values of ΔP^v_{sat}, but when compared to the low P^v_{sat} (due to the low CO_2-H_2O abundances), this leads to a large relative effect on ΔP^v_{sat} (i.e., 10–40%; Fig. 6c).

Ocean island basalts

Basalts related to hotspots have an f_{O_2} range that straddles the SS^{min} calculated for the reference basalt used in this study [e.g., ΔFMQ+1 to +3 using global compilations; Moussallam et al. (2019); Cottrell et al. (2021)] and can have much higher S-eq contents than MORB glasses (e.g., Ding and Dasgupta 2018). Hence S could significantly influence calculated values of P^v_{sat}. The broad positive correlation between S and f_{O_2} seen in Figure 6e (and also 6h for Marianas samples) is caused by progressive reduction during S degassing (e.g., Anderson and Wright 1972; Carmichael and Ghiorso 1986; Burgisser and Scaillet 2007; Métrich et al. 2009; Gaillard et al. 2011, 2015; Kelley and Cottrell 2012; Moussallam et al. 2014, 2016; Brounce et al. 2017). Due to the f_{O_2} range of the submarine matrix glasses from Mauna Kea and melt inclusion data from Kīlauea, and their low-H_2O and -CO_2 concentrations (<240 ppm CO_2-eq and $\leq$0.81 wt% H_2O-eq; Fig. 6d), the calculated ΔP^v_{sat} is up to 150 bars (the yellow symbols in Fig. 6f). Both these factors enhance the effect of the SS^{min} on calculated P^v_{sat}, leading to relatively large absolute and relative differences in P^v_{sat} (i.e., $\Delta P^v_{sat} = 30$–120 bars such that the relative effect is 20–60%; Figs. 6e and 6f).

Melt inclusions from Iceland also show the effect of elevated f_{O_2} on absolute differences in ΔP^v_{sat} (Fig. 6i), but a smaller effect on the relative increase in P^v_{sat} (i.e., ΔP^v_{sat} is <20% of the value of P^v_{sat}; Fig. 6f). This is due to the higher CO_2 concentrations of the Icelandic melt inclusions (Fig. 6d). Given the dominant influence of CO_2 on P^v_{sat} in these water-poor samples, P^v_{sat} for the Icelandic samples is generally higher than for the Hawaiian samples. Thus, while the increase in P^v_{sat} (which is approximately equal to ΔP^v_{sat}) is similar (several tens of bars) for the most oxidized samples from both groups as they approach the SS^{min} when normalized to the absolute value of P^v_{sat}, the percentage increase in P^v_{sat} for the most oxidized sample of the two sets of OIBs is smaller for the Icelandic samples (20% rather than 60%). This is likely a general rule: although ΔP^v_{sat} is similar in deeply trapped melt inclusions that have not experienced significant degassing of CO_2 and H_2O relative to shallowly trapped melt inclusions/matrix glasses, the relative effect on ΔP^v_{sat} will be lower in deep, undegassed melts than in shallow, degassed melts (Fig. 6f).

Arc basalts

Melt inclusion and matrix glasses from basalts from the Marianas arc have f_{O_2} values on the low f_{O_2}-limb of the SS^{min} (Fig. 6h). Thus, similar to OIB-related melt inclusions, arc-related melt inclusions with sufficient S also show higher calculated values of ΔP^v_{sat} with increasing f_{O_2} as S-bearing species contribute progressively more to the value of P^v_{sat}. For the Marianas samples, H_2O and CO_2 positively correlate, and the CO_2 concentrations are

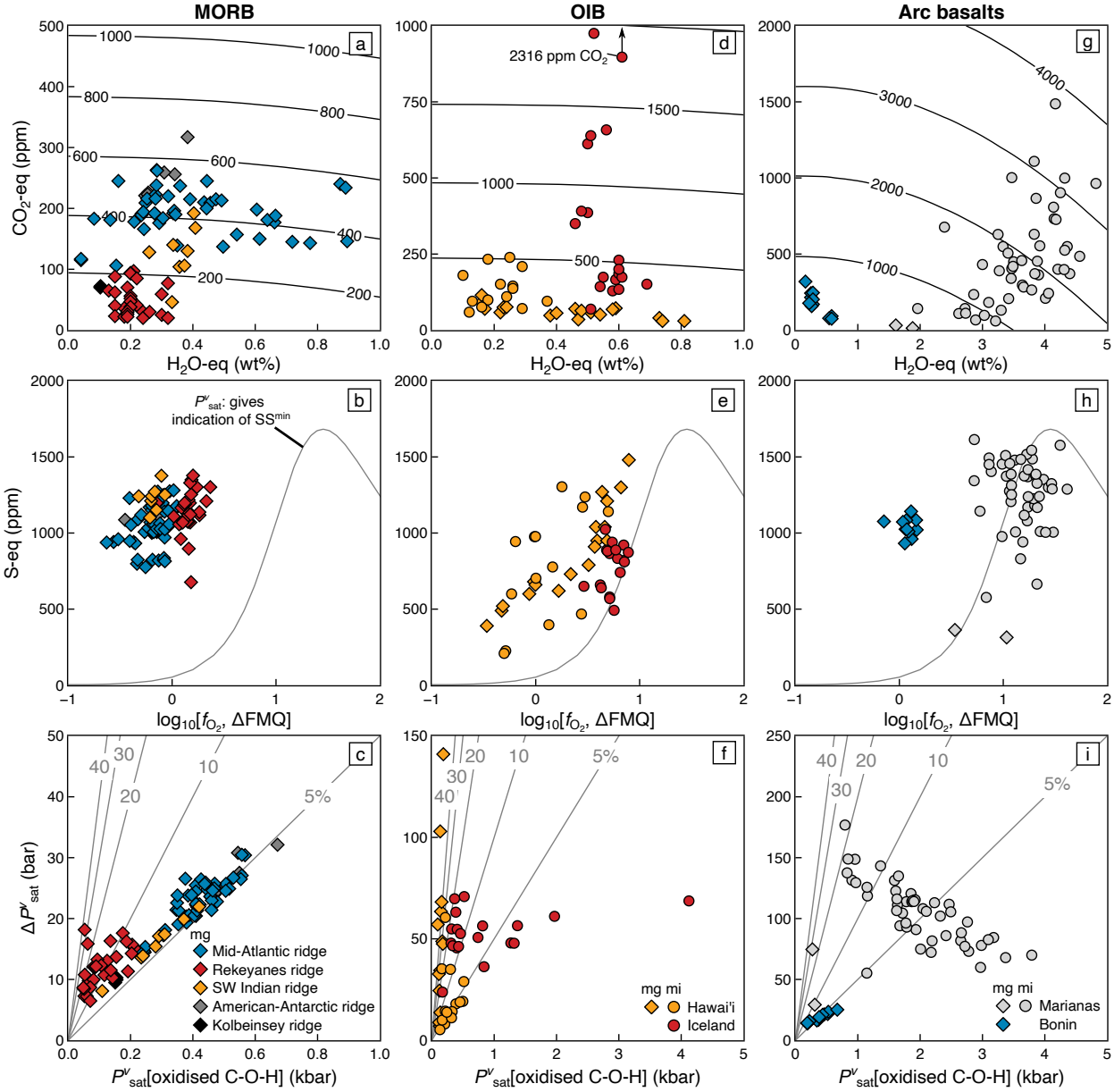

FIGURE 6. Melt inclusion (mi, circles) and matrix glass (mg, diamonds) data for natural terrestrial samples from the literature, for which H_2O-eq, CO_2-eq, and S-eq and Fe^{3+}/Fe_T have been measured. (**a**–**c**) Mid Ocean Ridge Basalts (MORB): American-Antarctic Ridge = gray, Kolbeinsey Ridge = black, Mid-Atlantic Ridge = blue, Reykjanes Ridge = red, and South-West Indian Ridge = yellow; (**d**–**f**) Ocean Island Basalts (OIB): Hawai'i = yellow, and Iceland = red; and (**g**–**i**) arc basalts: Bonin = blue and Marianas = gray. (**a**, **d**, **g**) Measured CO_2-eq vs. H_2O-eq, with calculated isobars (in kbar) using case 1; and (**b**, **e**, **h**) measured S-eq vs. calculated f_{O_2} from measured Fe^{3+}/Fe_T, with an indication of the location of the SS^{min} using case 4 (S-O, 1000 ppm S-eq) in gray—it does not use the *y*-axis values and is there to show the f_{O_2}-position of the SS^{min}. (**c**, **f**, **i**) ΔP^{v}_{sat} vs. P^{v}_{sat} (using case 1: only oxidized C-O-H species), where ΔP^{v}_{sat} is the difference between case 3 (all C-O-H-S species; calculated using measured volatile contents and melt composition at 1200 °C) and case 1. The gray solid lines show relative % errors (i.e., $100\cdot[\Delta P^{v}_{sat}/P^{v}_{sat}]$) as labeled on the line. None of the melt compositions are supersaturated with respect to graphite, sulfide, or anhydrite. MORB matrix glasses are from American-Antarctic Ridge (Le Voyer et al. 2019), Kolbeinsey Ridge (Le Voyer et al. 2019), Mid-Atlantic Ridge (Le Voyer et al. 2019), Reykjanes Ridge (Nichols et al. 2002; Shorttle et al. 2015), and South-West Indian Ridge (Wang et al. 2021). Hawai'i melt inclusions are from Kīlauea (Moussallam et al. 2016) and matrix glasses are from Mauna Kea (Brounce et al. 2017). Iceland melt inclusions are from 1783–1784 Laki eruption (Hartley et al. 2015, 2017). Marianas melt inclusions are from Agrigan, Sarigan, and Alamagan volcanoes (Kelley and Cottrell 2012; Brounce et al. 2014) and matrix glasses are from the Mariana Trough and Fina Nagu (Brounce et al. 2014, 2016). Bonin matrix glasses are forearc basalts (Brounce et al. 2021; Coulthard et al. 2022). (Color online.)

high (Fig. 6g). For the melt inclusions, this results in the negative correlation of ΔP^{v}_{sat} with P^{v}_{sat} (Fig. 6i), reflecting the larger impact of S in the f_{O_2} range of arc magmas when H_2O and CO_2 contents are lowest. The matrix glasses do not show this trend because of their lower f_{O_2} and S (Figs. 6h and 6i). However, as also observed for OIBs, ΔP^{v}_{sat} is small relative to P^{v}_{sat} (relative effect on $\Delta P^{v}_{sat} \lesssim$ 20%, and mostly $\lesssim$ 10% for melt inclusions and matrix glasses from the Marianas samples; Fig. 6i) due to their overall high-CO_2 and -H_2O contents (Fig. 6g). Conversely, matrix glasses from the Bonin islands are too reduced for the SS^{min} to be important and therefore ΔP^{v}_{sat} is small (blue diamonds in Figs. 6h and 6i).

Lunar and martian basalts

Lunar and martian basalts formed and evolved under significantly reduced conditions relative to modern terrestrial settings, with an f_{O_2} range of $-5 \lesssim \Delta FMQ \lesssim -1$ (e.g., Shearer et al. 2006; Gaillard et al. 2021). As was pointed out above and by others (e.g., Scaillet and Pichavant 2004; Wetzel et al. 2015; Newcombe et al. 2017; Lo et al. 2021) not including the effects of f_{O_2} as an independent variable can lead to significant underestimations in P^{v}_{sat} at the low f_{O_2}. This is experienced by lunar and martian magmas due to the low solubility of reduced C-O-H species. Calculated P^{v}_{sat} is very sensitive to f_{O_2} in this f_{O_2} range, so accurate f_{O_2} estimates are vital for accurate P^{v}_{sat} estimates. Additionally, primary magmas from both bodies are estimated to contain non-negligible S (up to ~900 and ~3500 ppm S-eq for the Moon and Mars, respectively, e.g., Gaillard et al. 2021). However, given that the SS^{min} is only likely to be observed at more oxidizing conditions than those prevalent on the Moon or Mars, this effect is unlikely to be important. However, the low-H and -C concentrations of these magmas ($\lesssim$1500 ppm H_2O and $\lesssim$6 ppm CO_2 in lunar melt inclusions; $\lesssim$2000 ppm H_2O and $\lesssim$200 ppm CO_2 in martian magmas; Gaillard et al. 2021; Saal and Hauri 2021) suggest that S-bearing species could be an important component of the vapor released by lunar and martian magmas on ascent.

Usui et al. (2012) and Saal and Hauri (2021) analyzed CO_2-eq, H_2O-eq, and S-eq in melt inclusions and matrix glasses from martian and lunar samples (Figs. 7a and 7b), but no measurements or estimates of f_{O_2} are available for the glasses. Hence, we calculated P^{v}_{sat} values for each glass at f_{O_2} values spanning the range $-5 < \Delta FMQ < -1$ at 1200 °C (e.g., Herd et al. 2002; Shearer et al. 2006), which generates the curves shown in Figures 7c and 7d (each curve is for a single analyzed glass). The range in ΔP^{v}_{sat} from Figures 7c and 7d are shown as vertical lines in Figures 7e and 7f. For both lunar and martian samples, H_2O-eq < 0.2 wt% but the lunar samples have low CO_2-eq (<16 ppm) and S-eq (900

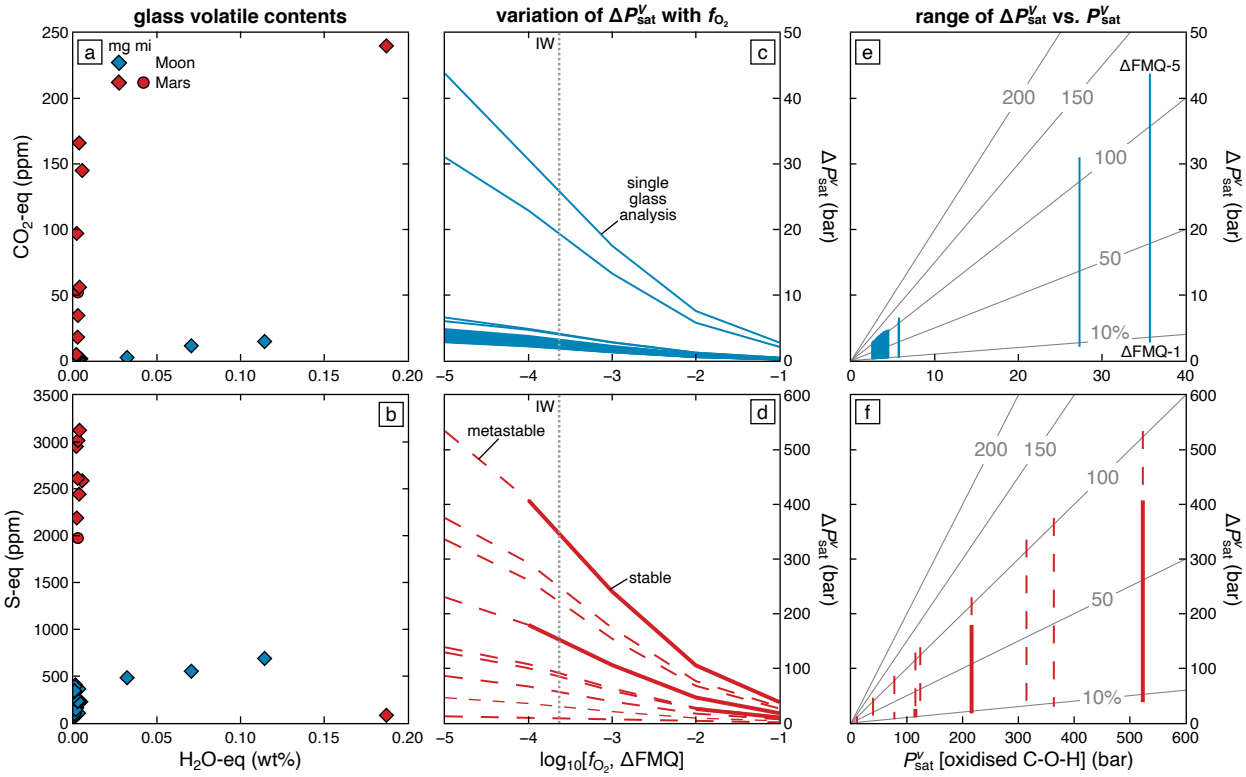

Figure 7. Melt inclusion (mi, circles) and matrix glass (mg, diamonds) data for natural lunar (blue; Saal and Hauri 2021) and martian (red; Usui et al. 2012) samples from the literature, where H_2O-eq, CO_2-eq, and S-eq have been measured. Measured H_2O-eq against (**a**) CO_2-eq and (**b**) S-eq. ΔP^{v}_{sat} (difference between cases 3 and 1) calculated for each measured glass composition at 1200 °C against (**c**–**d**) $\log[f_{O_2}]$ and (**e**–**f**) P^{v}_{sat} (case 1), where for the solid region of each curve the melt is not supersaturated with respect to graphite, sulfide, or anhydrite (labeled "stable") but the dashed regions indicate the melt is supersaturated with respect to sulfide and/or graphite (labeled "metastable"; for these conditions the melt is never supersaturated with respect to anhydrite). The extent of each vertical line in **e**–**f** corresponds to the vertical range of a single curve in **c**–**d**. Lunar samples are shown in **c** and **e** and martian samples in **d** and **f**. In **e**–**f**, the gray solid lines show relative % errors as labeled on the line (see Fig. 6). (Color online.)

ppm) relative to the martian samples (<240 ppm CO_2-eq and <3500 ppm S-eq) (Figs. 7a and 7b). The lunar samples record low pressures (i.e., P^v_{sat} < 50 bars; Figs. 7c and 7e), and excluding S and reduced C-O-H species leads to underestimates of P^v_{sat} by up to 130% at ΔFMQ-5 (Fig. 7e). The martian samples record much higher pressures but the relative change in P^v_{sat} by including S and reduced C-O-H species is still large, up to 100% at ΔFMQ-5 (Fig. 6p). Based on our calculations, neither sulfide- nor graphite-saturation is indicated for lunar samples but most of our calculations for martian samples suggest sulfide (and sometimes graphite) saturation are reached; in such cases, the calculated equilibrium between melt and vapor is metastable, indicated by dashed curves (Figs. 7d and 7f). Care must be taken applying these results as the carbonate solubility used may not be applicable to lunar and martian compositions (although sulfide and sulfate solubilities are appropriate).

Our results are in general agreement with Wetzel et al. (2015) and Newcombe et al. (2017), who calculated P^v_{sat} for lunar glasses including reduced C-O-H species in the system: OH^-, H_2O_{mol}, and $Fe(CO)_5$ in the melt and CO and H_2O in the vapor (Wetzel et al. 2015); and OH^-, CO_3^{2-}, and CO in the melt and H_2, H_2O, CO, and CO_2 in the vapor (Newcombe et al. 2017). In general, this highlights that modeling lunar and martian magmas requires the inclusion of both reduced and oxidized C-O-H-S species (Newcombe et al. 2017), and this will have implications for where magmas are stored and eruption dynamics (e.g., Lo et al. 2021). Also, these calculations of P^v_{sat} are highly dependent on the f_{O_2} used to calculate P^v_{sat}, and accurate f_{O_2} estimates are needed for accurate P^v_{sat} estimates (e.g., Figs. 7e and 7f). However, quantitative results are sensitive to the solubilities in the melt of the reduced species, and more work is necessary to ensure their accuracy, but the trends shown here are robust.

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